DELIVERABLE 1 · AI OPPORTUNITY REPORT

Turning customer feedback into safe, fast, auditable action

Risk-gated review and ticket triage for Revio (hospitality reputation) and TenantEase (residential operations)

PLATFORM	DEPLOYMENT	STATUS	REVISION
Revio · TenantEase	Vercel Serverless Edge	Live & verified	v2.0 · November 2026

35–60%	< 15 min	$53K–$163K	< 0.8%
of routine praise automated at maturity (target)	escalation time for severe defects (target)	annual triage savings per 4,000-review/month portfolio (modeled range)	automated-misclassification rate (target, to be audited)

Executive summary

Cynbit's **Revio** (multi-location reputation management) and **TenantEase** (residential property operations) ingest thousands of unstructured reviews, resident surveys and maintenance tickets every month. Manual triage is slow and expensive. Naive AI automation is fast but unsafe: it can publish a cheerful thank-you under a sarcastic review or a food-poisoning complaint.

This report proposes a **risk-gated Feedback Intelligence & Triage Engine** that separates *classification* from *action*. Clear, in-domain praise is acknowledged automatically. Ambiguous, mixed-signal and high-risk feedback goes to people, together with the exact language cues that produced the score. Uncertainty is treated as an operational signal, not statistical noise.

A working version is live (Deliverable 2) and its engineering is documented in Deliverable 3. The business case below is deliberately presented as a **range** with every assumption visible, so Cynbit can replace our planning inputs with its own numbers during a 4-week shadow pilot.

> **How to read the numbers.** [MEASURED] observed on the live proof of concept. [MODELED] derived from stated assumptions; replace with customer data. [TARGET] a design goal to be validated in Phase 1 shadow mode. Financial figures are always MODELED, never claimed as measured results.

1. The business problem

Revio: reputation at multi-location scale

- **Response latency.** Brand managers cannot read and answer reviews across dozens of locations within minutes; a typical manual lag is 12–24 hours (planning assumption). Response speed and rate are widely believed to influence local-search visibility on Google and Yelp.
- **The sarcasm hazard.** Keyword bots read "What a treat to be yelled at by the manager! Just what my anniversary needed." as positive and publish "We love delighting our guests!" under it. One such reply can become a public-relations incident.
- **Defect invisibility.** Food poisoning, cockroaches and billing fraud sit in the same stream as routine feedback and are not routed to an operator quickly.

TenantEase: operations and resident retention

- **Critical maintenance delays.** HVAC failure in extreme heat, water leaks and mold arrive through the same portal as general questions, delaying dispatch and risking property damage and code violations.
- **Predictable churn blindspots.** Residents rarely leave without warning. "Great amenities and location, but the hallway noise and elevator breakdowns are unbearable" signals non-renewal months ahead, yet is easily missed by stretched

property teams.

2. The opportunity: risk-gated triage

The core idea is to decouple **what the model believes** (a probability) from **what the business does** (a routing decision with an owner and a service-level target). Four rules are applied in strict precedence: ambiguity first, then severe defects, then validated praise, then unvalidated-domain quarantine.

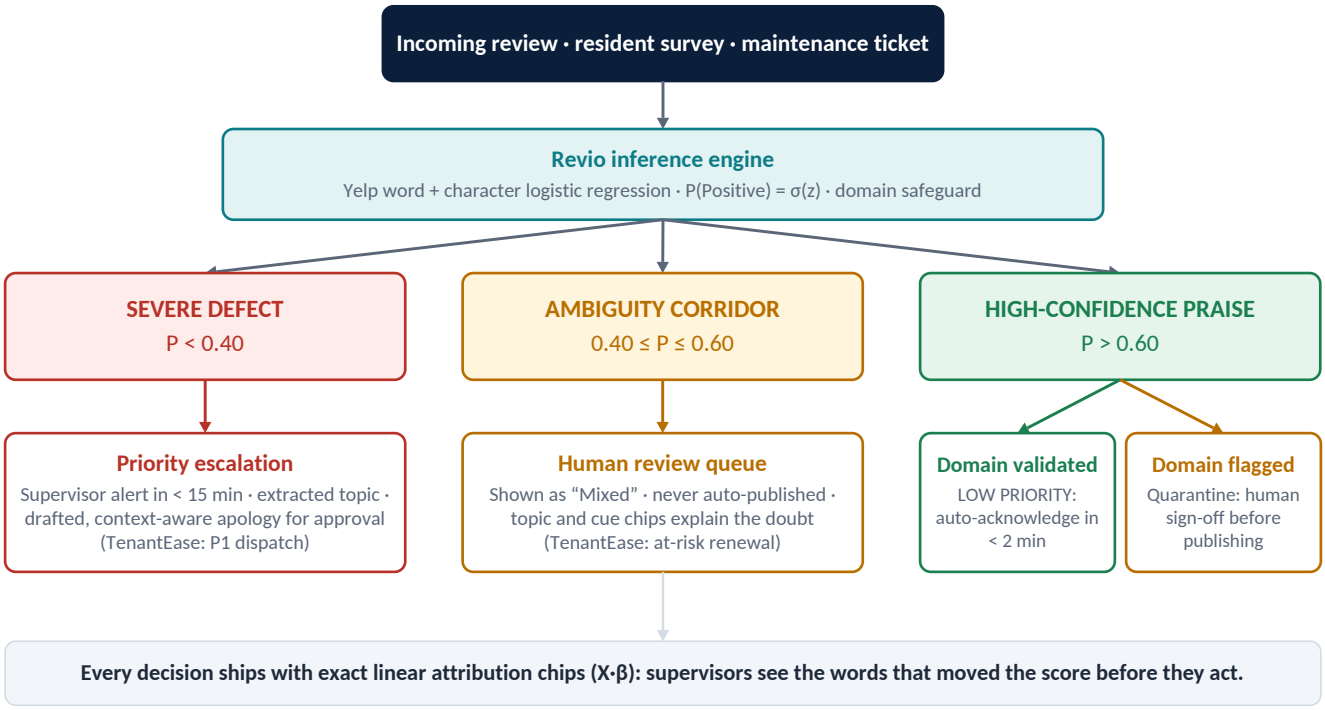


Figure 1. Operational triage architecture. Probabilities are model outputs; actions, owners and SLAs are business policy.

Lane	Condition	System action	Human role	Target SLA
Priority escalation	P < 0.40	ESCALATE (Negative / High Risk); topic + cue chips; drafted apology	Supervisor approves / resolves	< 15 min
Ambiguity corridor	0.40 ≤ P ≤ 0.60	ESCALATE (Ambiguity / Low Confidence); shown as "Mixed"; no auto-publish	Editor decides	Same business day
Validated praise	P > 0.60 and domain validated	LOW PRIORITY / NO HUMAN REVIEW REQUIRED; acknowledge automatically (staged gate, §6)	Sampled audit	< 2 min
Domain quarantine	P > 0.60 and domain not validated	ESCALATE (Unvalidated Domain); publishing blocked	Human sign-off	Same business day

Table 1. Triage lanes. SLAs are proposed targets.

Four design principles

- **Ambiguity is a feature.** Reviews such as "Delicious food, but agonizingly slow service" are flagged and sent to an editor instead of being forced into a binary label.
- **Severe grievances are intercepted deterministically.** The engine surfaces the cues behind a low score ("leak", "poisoning", "overcharged", "cockroach") and categorizes the friction as Maintenance, Billing, Staff or Food Quality.
- **Automation is earned, not assumed.** Only unambiguous, domain-validated praise is auto-acknowledged, and the confidence gate is tightened for the pilot (Section 6).
- **Text outranks metadata.** If a request says "restaurant" but the text describes a mobile-network outage, the engine overrides the label and quarantines the item.

3. Financial impact model

Reference portfolios: a **hospitality group** with 50 locations and 4,000 reviews a month, and a **property group** with 5,000 residential units and 4,000 tickets and survey comments a month. The two are alternative profiles and their savings are not additive. Because the earlier draft mixed two incompatible labor-cost bases, the model below presents both explicitly.

Assumptions

Parameter	Value	Basis
Monthly volume	4,000 items	Reference portfolio
Manual handling time	4 min / item	Read, classify, route and draft; planning assumption
Fully burdened labor rate	$25 / hour	Scenario A (bottom-up)
Handling time after triage	2 min / escalated item	Cue chips and drafted responses shorten review; auto-acknowledged items need no handling
Cloud and tooling	≈ $50 / month	Serverless inference; negligible at this volume
Automation share	35% · 50% · 60%	Phase 2 target, mid-point, Phase 3 target

Scenario A: conservative, bottom-up labor model [MODELED]

Automation share	Staff hours / month	Monthly cost	Monthly saving	Reduction	Annual saving
Baseline (manual)	266.7 h	$6,667	—	—	—
35% automated	86.7 h	$2,217	$4,450	66.8%	$53,400
50% automated	66.7 h	$1,717	$4,950	74.3%	$59,400
60% automated	53.3 h	$1,383	$5,283	79.3%	$63,400

Scenario B: fully-loaded cost per item [MODELED]

If Cynbit's loaded cost per manually handled item is **$4.50** (including management oversight, drafting, approvals and tooling) and the blended cost with the engine is **$1.10**, monthly cost falls from $18,000 to $4,400: a saving of **$13,600 per month ($163,200 per year, a 75.6% reduction)**. This matches the earlier headline and should be adopted only if Cynbit's own finance data supports a $4.50 baseline.

Operational and retention effects

Metric	Manual	Naive single-model AI	Risk-gated engine	Status
Response lag, positive praise	≈ 18 h	Instant (< 1 min)	< 2 min	[TARGET]
Escalation time, severe defect	≈ 8.5 h	Uncontrolled	< 15 min	[TARGET]
Automated misclassification / PR error rate	n/a	8.5–12.0% (planning assumption)	< 0.8%	[TARGET]
Resident churn (TenantEase)	14% baseline	14% baseline	12.5% (−1.5 pp) to 11.5% (−2.5 pp)	[MODELED]

Resident retention. Average turnover cost is $3,850 per unit (cleaning, repainting, marketing, vacancy). In a 5,000-unit portfolio, a 1.5-point churn reduction avoids 75 vacancies, preserving **$288,750** a year; a 2.5-point reduction avoids 125 vacancies, preserving **$481,250**. Per 1,000 units this is $57,750 to $96,250. These are hypotheses to test: the engine is trained on restaurant and hotel reviews, so TenantEase effects must be validated in shadow mode before they are claimed.

> **Evidence note.** The misclassification target is a design goal. Phase 1 measures it against human ground truth on Cynbit's own history before any automation is switched on.

4. Product workflows

Revio: intelligent brand-reputation automation

Incoming signal	Engine output	Revio action
Positive, P > 0.60, domain validated	LOW PRIORITY / NO HUMAN REVIEW REQUIRED	Acknowledge and prompt for a referral (staged gate; < 2 min)
Mixed, 0.40 ≤ P ≤ 0.60	ESCALATE (Ambiguity / Low Confidence)	Inbox "Needs review" with ambiguity warning
Negative, P < 0.40	ESCALATE (Negative / High Risk) + topic	Inbox with extracted topic (e.g. Food Quality); manager approves a drafted apology

Feature attribution in action. For a cold-food complaint the engine exposes the contributing cues (for example "cold food" and "undercooked", shown as illustrative weights of −0.65 and −0.52), so the drafted reply can acknowledge the specific issue rather than send a generic apology.

TenantEase: predictive maintenance and churn prevention

Incoming signal	Engine output	TenantEase action
Severe defect, P < 0.40	Topic: Maintenance & Facility or Billing	Dispatch P1 emergency alert; notify on-site supervisor
Ambiguous survey, 0.40 ≤ P ≤ 0.60	Flag: at-risk renewal	Queue for resident-relations outreach, ideally ~90 days before lease expiry

Example. "The water heater is leaking onto the bedroom carpet and nobody answered the emergency line" scores P(Positive) = 0.008, extracts a facility-defect topic and bypasses clerical queues to alert emergency maintenance.

5. Risk management and governance

Four built-in controls

- **No silent model failures.** If the selected model artifact is missing or corrupted the API returns HTTP 503; it never falls back silently to a model with different decision thresholds. Rollback to the champion model is a deliberate operator action.
- **Text dominance over metadata.** Client-supplied domain labels are untrusted and are overridden when the text disagrees.
- **Precise workflow language.** Automated outcomes read LOW PRIORITY / NO HUMAN REVIEW REQUIRED, not a vague "auto-acknowledge", so oversight is always visible.
- **Transparent explainability.** Every inference carries exact linear attributions ($X_j \cdot \beta_j$) that supervisors can inspect before approving or editing a response.

Residual risks and mitigations

Risk	Why it matters	Mitigation
Sarcasm and irony	A lexical n-gram model can read positive words inside a negative message; the corridor catches many, not all, such cases.	Auto-acknowledge only at high confidence in the pilot (P > 0.85); keep sarcasm cases in the benchmark; sampled human audit of auto-acknowledged items.
Domain shift to TenantEase	Training data is Yelp restaurant and hotel reviews; maintenance and resident language differs.	Shadow-evaluate on TenantEase history first; withhold TenantEase auto-actions until concordance is demonstrated; quarterly retraining on supervisor corrections.
Probability drift and calibration	Corridor and gates assume P behaves like a calibrated probability.	Reliability audit before go-live; monitor the distribution of P weekly; recalibrate or retrain on drift.
Automation bias	Reviewers may over-trust a confident score.	Show probability, distance from the boundary and cue chips; require explicit "Mark as reviewed" with an audit trail.

6. Phased rollout

Rule 3 classifies praise at P > 0.60. Publishing automatically is a separate, stricter, configurable gate that starts high and is lowered only when audit evidence supports it. This replaces the earlier draft, which mentioned 0.60, 0.85 and 0.55 for the same decision.

Phase	Scope	Auto-acknowledge gate	Exit criteria (proposed)
1. Shadow Weeks 1–4	Deploy PoC; score historical and live reviews in parallel with manual work; measure concordance with human decisions.	None	Concordance report; escalation lanes capture ≥ 95% of human-labeled severe defects; zero downtime.
2. Pilot Weeks 5–8	Enable gated acknowledgment; route all else to the Review Inbox; supervisors approve negative-reply drafts. Target 35% automation.	P > 0.85 and domain validated	< 1% of audited auto-acknowledgments overturned; median first response < 2 min.
3. Production Weeks 9–12	Connect Google Business Profile, Yelp Fusion, TripAdvisor and TenantEase; active-learning loop; resident-churn scoring. Target 60% automation.	Lower toward P > 0.60 only if Phase 2 audit passes	Quarterly retraining on supervisor corrections; monitored drift; audited automation rate.

7. Strategic conclusion

The Revio engine gives Cynbit a defensible, enterprise-grade capability: **autonomous speed where language is clear, and transparent human oversight where risk exists**. It is inexpensive to run, explainable on every decision and already live for evaluation.

> **Decisions requested.** (1) Approve a 4-week Phase 1 shadow pilot. (2) Provide 3 months of historical reviews or tickets with human outcomes for concordance testing. (3) Nominate pilot locations and a TenantEase property. (4) Supply Cynbit's own cost-per-item and turnover figures to replace the model assumptions in Section 3.